

MATH 3551 A — Abstract Algebra I

Additional Practice Midterm 1 — Version B

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For practice — Midterm 1 is Monday, October 12, 2026 • Time limit: 50 minutes • Total: 100 points

NAME (please print): _____

Directions. This exam is **closed book**: no textbook, no notes, no calculators, no phones, no other devices, and no assistance of any kind. Write your answers on the exam itself; use the backs of pages if you need more room, and say clearly when you have done so. *Show your work and name the theorems you use* — “by Lagrange’s Theorem, $|H|$ divides 24” earns nearly all of the credit, an unjustified correct number earns little. Point values are marked on each problem. Good luck.

About Version B. This additional practice exam was generated with GPT-5.6 Terra in Codex. It targets about thirty minutes of work in this fifty-minute period. Please report any error, ambiguity, out-of-coverage item, or unexpectedly time-consuming question to Professor Dupuy.

Problem 1. (30 points) **True or false; prove or disprove.**

Circle **T** or **F**, and then give a one-line reason: a theorem, a short computation, or an explicit counterexample. Five points each: two for the circle, three for the reason.

- (a) **T F** The congruence $14x \equiv 8 \pmod{20}$ has a solution.
- (b) **T F** If H and K are subgroups of a group G , then $H \cup K$ is a subgroup of G .
- (c) **T F** In the dihedral group D_{14} of order 14, the product of any two reflections is a reflection.
- (d) **T F** The determinant map $\det: \mathrm{GL}_2(\mathbb{R}) \rightarrow \mathbb{R}^\times$ is a group homomorphism.
- (e) **T F** The additive group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ is cyclic.
- (f) **T F** If $\varphi: G \rightarrow H$ is a group homomorphism and $\varphi(g) = 1_H$, then $g = 1_G$.

Problem 2. (25 points) **A reduction homomorphism.**

Work additively. Define a map

$$\pi: \mathbb{Z}/24\mathbb{Z} \longrightarrow \mathbb{Z}/8\mathbb{Z}, \quad \pi(\bar{a}) = \bar{a}.$$

- (a) (4 points) Prove that π is well defined.
- (b) (4 points) Prove that π is a group homomorphism.
- (c) (5 points) Determine $\ker \pi$ and $\operatorname{im} \pi$.
- (d) (4 points) Let $K = \langle \bar{6} \rangle \leq \mathbb{Z}/24\mathbb{Z}$. Determine $\pi(K)$ by listing its elements in $\mathbb{Z}/8\mathbb{Z}$.
- (e) (8 points) Determine every $\bar{a} \in \mathbb{Z}/24\mathbb{Z}$ for which $\pi(\bar{a})$ generates $\mathbb{Z}/8\mathbb{Z}$.

Problem 3. (25 points) **A matrix group over $\mathbb{F}_2$.**

All matrix entries and calculations in this problem are in $\mathbb{F}_2$. In $\text{GL}_2(\mathbb{F}_2)$ let

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad H = \langle A, B \rangle.$$

- (a) (5 points) Compute A^2 and determine the order of A .
- (b) (5 points) Compute BAB and compare it with A^{-1} .
- (c) (10 points) List the elements of H as matrices, and prove that $H = \text{GL}_2(\mathbb{F}_2)$.
- (d) (5 points) Determine the order of AB .

Problem 4. (20 points) **A subgroup of $\mathbb{Z} \times \mathbb{Z}$.**

Let $G = \mathbb{Z} \times \mathbb{Z}$ under componentwise addition, and let

$$H = \{(a, b) \in G : 3 \mid (a + b)\}.$$

- (a) (5 points) Prove that H is a subgroup of G .
- (b) (9 points) Prove that $H = \langle (1, -1), (0, 3) \rangle$.
- (c) (6 points) Prove that H is not cyclic.